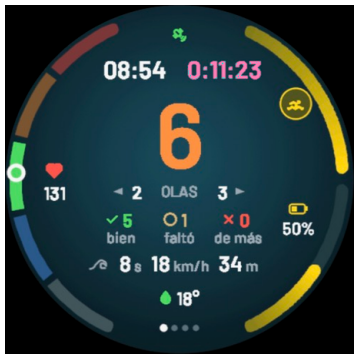
Surf Dawn
User manual

The watch face, Forecast and Session for Garmin watches







24 October 2026

The three apps

Surf Dawn is three apps for your Garmin watch that work together. They share the spots, the colours and the icons:

App	What it's for	How to open it
The face	See at a glance how the sea is at your spot: waves, wind, tide, water, air, sun and battery	It is the watch face
Forecast	See the day hour by hour and which hours are the best; in the morning, an alert with the day's best spot	Touch and hold the face, or from the app list
Session	Record your session, count your waves, find your way back to the spot and save everything to Garmin Connect	Touch and hold the waves subdial, or from the app list

Getting started

1. **Install the three apps** (see Install).
2. **Put the face on:** hold **MENU** (the middle button on the left) → **Watch Face** → choose **Surf Dawn** and press **START**.
3. **Keep your phone nearby** with the Garmin Connect app. Your spot's data arrives in a few minutes; until then the face says "LOADING".
4. **Choose your units and your background** in each app: on the face from Customize, and in Forecast and Session by holding **UP**. They come set to metres, km/h and the Sea background.
5. **Load your boards** on the phone: Connect IQ app → Surf Dawn Session → Settings → Boards. Session doesn't ship with any boards; if you don't load them, sessions are recorded without a board.
6. **Tell it how you surf:** in Session, by holding **UP**: **Stance** (Goofy by default, or Regular) and **Watch on** (your wrist, left by default). With your stance, each wave is logged as frontside or backside.
7. **Check the forecast:** touch and hold the face. The three best hours of the day come marked.
8. **Before you go in the water,** open Session, wait for the satellite at the bottom left to turn **green** and press **START**.
9. **In the water you don't need to touch anything.** If you want, press **BACK twice** after each wave to mark it: that is how the watch learns to count your waves. With DOWN you go from the session to the compass pointing back to the spot, the forecast and your waves (UP goes back).
10. **When you come out: START → Save → START**. Look at the summary, give it stars and tags, and that's it: it is saved in Garmin Connect.

A tip for your watch: under **Display → During activity**, choose **Always on**, so you can see the session without touching anything.

The buttons

Supported watches have five buttons: on the left **LIGHT** (top), **UP** (middle; held down it is **MENU**) and **DOWN** (bottom); on the right **START** (top) and **BACK** (bottom).

Button	Forecast	Session, before entering	Session, in the water
START	Request the data again	Start the session	Open the menu (Save, Pause, Waves…)
BACK	Exit	Exit	Nothing; **twice** marks a wave
UP	Choose the spot	See the day's conditions	Previous page
DOWN	Choose the spot	Menu: routines and Settings	Next page
UP held (MENU)	Settings	Settings	Settings
LIGHT held	Watch controls: light, inReach, SOS	same	same

The face doesn't use buttons: on watches with a touchscreen, touching and holding it opens Forecast or Session (Gestures).

The Surf Dawn watch face

A watch face with hands (or with the time in digits) that shows you the sea at the spot where you are: waves, wind, tide, water and air temperature, the sun and the battery. It picks the nearest spot on its own and updates itself every half hour (every 15 minutes between 5 and 10 in the morning).

It has four backgrounds: **Sea**, **Night**, **Graphite** and **Forest**.



And the same information with the time in digits:



What you see

Where	What it is	What it's for
Top	The time of the next sunrise (yellow sun) and the next sunset (orange sun)	Planning your first and last session
Top middle	The battery: green (over 50 %), yellow (20 to 50 %) or red (under 20 %)	
Circle with the droplet	The sea temperature	Choosing your wetsuit
Circle with the weather	The air temperature and the weather icon (sun, clouds, rain...). The **colour of the number is the sun (UV)**: white with no sun, green low, yellow moderate, orange high, red very high, violet extreme	Knowing whether you need sunscreen

Where	What it is	What it's for
Left subdial	The **waves**: big height (metres or feet) with the period below ("10s"). The little marker on the edge shows **where they come from**. Its colour tells you whether they reach your beach (see below)	Seeing the size and whether the swell arrives
Right subdial	The **wind**: speed (km/h or knots) and where it blows from ("SE"). The colour tells you whether it is offshore, cross-shore or onshore	Seeing whether the sea is clean
Date capsule	The day and the date. It moves out of the way by itself so the hands don't cover it	
Bottom curve	The **tide** for the next hours: the turquoise dots are high tides, the sand-coloured ones low tides, with their time. The little white triangle is now, with the current height below it	Knowing whether it is rising or falling and when it turns
Curved band	The name of the **spot**. A little turquoise dot on the left means you chose it yourself (fixed)	Knowing which spot the data is for
Near the centre (in Digital, under the battery)	Only when something is up: "LOADING" (the spot's data hasn't arrived yet), "4 H AGO" in red (the data is hours old and the hourly forecast that was downloaded has already run out: the numbers turn grey; "2 D AGO" if it is days) or a crossed-out phone (the watch isn't connected to the phone)	Knowing whether the numbers are up to date

The colours

Colour	Waves (marker and period)	Wind (marker and letters)
Green	They come in head-on to the beach	Offshore: blows from the land, the sea stays clean
Yellow	They come in at an angle	Cross-shore: blows from the side
Red	They hit it from the side or from behind: the beach is blocked	Onshore: blows from the sea, chopping it up

Each spot knows which way its beach faces; that is what picks the colour.

Time in digits

In Customize → Time → **Digital**, the time goes big in the middle. Waves and wind move up to the top, the battery sits between the two, and the water and air go on either side of the date. The tide stays at the bottom and sunrise and sunset go at the very top.

Always on



When the screen is at rest (AMOLED displays with "always on" enabled on the watch), the face switches to a dark grey version that barely uses any battery: date, time and three rows with an icon: - **Waves:** height, period and where they come from. - **Wind:** speed and where it blows from. - **Tide:** the next one, with ▲ if it is high tide or ▼ if it is low tide, and its time.

Everything shifts a tiny bit every minute so the screen doesn't burn in. At rest the stale-data warning isn't shown.

Gestures: touch and hold

On watches with a touchscreen, touching and holding the face opens the other apps:

Where you hold	What opens
The waves subdial	Session, to record a session
The band with the spot name	The spot picker in Forecast. The spot you choose also carries over to the face
The top row (sunrise and sunset)	inReach messages, if your watch offers them for watch faces; otherwise, Forecast
Anywhere else	The day's Forecast

If Session isn't installed, the waves subdial opens Forecast. If Forecast isn't installed, holding the band moves on to the next spot in the same zone (and after the last one, to Automatic). On watches without a touchscreen, open Forecast and Session from the app list.

Customize

Hold **MENU** (the middle button on the left) → **Watch Face** → with Surf Dawn on screen, **START** → **Customize**:

Option	Values (defaults in bold)	What it changes
Spot	**Automatic** or a fixed spot	Automatic shows which one it is choosing ("Active · LA PLAGE"). Below are the 30 nearest spots from the zones the watch has, with their break type and distance. Choosing one makes it **fixed** until you go back to Automatic
Zones	The watch's zones	How many spots each one has and whether it came with the face or was downloaded. Choosing a downloaded one deletes it (it asks for confirmation), including the one in use (the spot goes back to Automatic); it downloads itself again when you are there
Time	**Hands**, Digital	See Time in digits
Background	**Sea**, Night, Graphite, Forest	The background colour. Graphite, nearly black, uses the least battery
Waves	**Total**, Swell	Total = the whole sea, the groundswell plus the wind waves (it is Windguru's "Wave" row). Swell = the groundswell only, with its period and direction
Height (waves and tide)	**Metres**, Feet	
Wind	**km/h**, Knots	

When you change spots, the data arrives in a few minutes; meanwhile you see "—" and "LOADING".

How it picks the spot

- Spots come in **zones** of up to 60 (Mar del Plata, Oahu's North Shore, Gold Coast, Mentawai…): over 2,000 spots in about 200 zones around the world. The face ships with Uruguay's, from Montevideo to Chuy: Canelones and Montevideo (Playa Honda, La Pantalla and the Costa de Oro, from El Pinar to Jaureguiberry), Maldonado (from Punta del Chileno to José Ignacio, with the Brava's numbered beach-access stops) and Rocha; **it downloads the rest by itself** when you arrive, Florianópolis and Praia do Rosa included: the first time in a new place it takes about 10 minutes (first it looks up which zones are there and then downloads yours) (you see "—" and "LOADING") and from then on it stays saved. It keeps up to 10 zones besides the built-in ones; when there is no room, it deletes the one you haven't used for the longest.
- It picks the **nearest one to where you are**, if it is less than 25 km away. It doesn't turn the GPS on: it uses the last location the watch has and checks it every 10 minutes. If you haven't done a GPS activity, that location is the one the phone passes to it and it can be far off (in Punta del Este it was once 19 km away): if the face shows a spot that isn't yours, open Session until the GPS turns green, and check that Garmin Connect has precise location on the phone.
- If you chose a **fixed** spot, it uses that one.

- If there is no spot nearby, it uses the **fallback spot**. By default it is La Plage. With the app installed from the store, you change it on the phone: Connect IQ app → Surf Dawn → Settings (name, latitude and longitude out at sea, and which way the beach faces, in degrees).
- If you chose, in Forecast, a spot from a zone the face doesn't have, it downloads the zone and then switches to that spot.
- Every update also downloads the data for the neighbouring spots in the zone (up to 8, within 25 km). That is why switching to a spot next door is instant. A distant one takes a few minutes: Garmin only allows background updates once every 5 minutes.
- Each update also brings the **hourly forecast** for the spot in use for the next 30 hours (waves, wind, temperature and sky). Without the phone, the face steps forward to the current hour: at 15:00 it shows what was forecast for 15:00, even if the last download was from the morning. The numbers only turn grey when those hours run out.
- To download new data the watch needs the phone nearby with the Garmin Connect app.

On other watches

- **MIP displays** (fēnix 7, Forerunner 955 and 255, fēnix 8 and 9 Pro Solar, Enduro 3): flat colours and no second hand. These watches' displays never turn off, so at rest you still see the full face.
- **Instinct Crossover AMOLED:** it has real hands, so the face draws neither its own hands nor the time in always-on mode, and it has no Time option.
- **Watches without a touchscreen** (Forerunner 255 and 255 Music, Instinct 3 AMOLED, Instinct Crossover AMOLED): they don't have the touch-and-hold gestures.

The full list is in Supported watches. Where each number comes from: Data and models.

Surf Dawn Forecast

Your spot's forecast for the day, hour by hour, on a single screen: waves, wind, tide, water and air, with the three best hours marked. It uses the same models as Windguru and the official tide tables. If you want, every morning it tells you which spot near you is the best one today (The morning alert).



Opening it

- From the Surf Dawn face: **press and hold the face** anywhere (outside the name band and the wave subdial).
- From the face, holding the **name band**: opens the spot picker directly.
- From the watch's app list: "Surf Dawn Forecast".

When it opens it requests the data right away (it takes a second or two, with your phone nearby).

How to read it

From top to bottom:

1. **The day and the spot:** "TODAY · SAT 10 OCT" and the spot's name.
2. **Daylight:** the arc at the top runs from sunrise to sunset and fills up to the current time, with a little sun at the current time. Below its two ends, the sunrise and sunset times.
3. **The waves:** one bar per hour, from 5 to 19 h. Taller = bigger waves. Above the bars, the height wherever it changes. On the left, the unit and the period for the day ("10s", or "9-12s" if it changes).
    - **The bar's colour** tells you whether the waves reach your beach: green head-on, yellow diagonal, red from the side or from behind.
    - **The little arrows** inside the bars (every two hours) point **where the waves are heading**.
    - Hours that have already passed look dimmer, and a white line with a little triangle marks the current hour.
4. **The best-hours band:** a box behind the three most favorable hours with daylight. See The best hours.
5. **The wind**, every two hours: an arrow showing where it blows toward and the speed, in green (offshore), yellow (cross-shore) or red (onshore).
6. **The tide:** the day's curve with the time of each high tide (turquoise dot) and low tide (sand dot), and a white dot at the current time.
7. **Water and air:** the current sea temperature (drop) and air temperature (thermometer); for tomorrow and the day after, the temperature at 8.
8. **The dots at the bottom:** which day you're on (today, tomorrow, the day after).

The best hours

The box marks the three consecutive hours, with daylight, that look best. For each hour, the watch looks at:

- **The size** (counts most): nothing at 30 cm, best from 1.8 m.
- **The period:** nothing at 5 s, best from 13 s.
- **The wind:** it doesn't bother you up to a certain point, which depends on where it blows from: up to 35 km/h if offshore, 12 km/h if cross-shore and 8 km/h if onshore. Stronger than that, the hour is worth less and less.
- **Whether the waves reach the beach:** head-on ones count in full, diagonal ones a bit less, blocked ones much less.

If the day has nothing worthwhile, the box doesn't appear. This calculation is the same for everyone; the stars based on your own sessions are in Session.

Changing the day

Swipe left to see tomorrow and the day after, and right to go back. After 19 h it opens straight on tomorrow.

On watches without a touchscreen (Forerunner 255, Instinct 3 AMOLED, Instinct Crossover AMOLED) only the first day is shown.

Choosing the spot

UP, **DOWN** or a tap on the screen open the spot picker.

- It opens on the **zone** you're in (for example "Maldonado" or "North Shore de Oahu"), with **Automatic** always at the top (it says which one is the nearest, or "nearest to where you are" if you're far from the spots; if it's the one in use, it says "active") and the spots from nearest to farthest. Each one shows its distance and a little arrow pointing to where it is (with north up).
- **BACK** shows the folders:
  - **The zone in use**, with how many spots it has.
  - **Saved zones**: the ones the watch has (the factory ones, which are Uruguay's, and the downloaded ones). Inside a downloaded one, the last row, **Remove this zone**, deletes it (it asks for confirmation), even the one in use: the picker goes back to Automatic and the zone downloads itself again when you're there.
  - **The whole world**: continent → country → region. Each region says how many spots it has and whether it's already saved or how much it weighs to download (2 or 3 KB). Choosing it downloads it and opens it.
- When you arrive somewhere new, the app downloads that zone on its own.
- When you pick a spot, its forecast is downloaded and **the Surf Dawn face switches to that same spot** (if it doesn't have that zone, it downloads it). Automatic goes back to picking the nearest one.
- Each app keeps its own zones (up to 10 besides the factory ones): when there's no room, the one you haven't used for the longest goes, and the ones you downloaded by hand go last.

Break type. The icon on each spot tells you how it breaks:

Icon	Type
The wave over a line of sand	Beach break (sand bottom)
The wave over three round stones	Point break (breaks against a point)
The wave over a row of teeth	Reef (rock or reef bottom)
A pin	Unknown

If you disagree with a spot's type, the last row of the picker, **Break type**, changes it: pick the spot and then Beach break, Point break, Reef or Unknown. It's marked "edited", and **As shipped** puts it back to the original. The change stays in Forecast only: Session and the face have their own.

The morning alert



Once you turn it on, every day at the time you choose (06:00 out of the box) the watch vibrates with **the best spot to surf today** among the ones near you: "Today: EL EMIR", and below it the waves and period in the day's best window ("1.4 m 6 s · 07-09 h") and the next two spots.

- **How it picks:** it compares the nearest spots of your zone (up to 6, within 25 km of where you were the last time you closed the app; if the watch didn't know where you were, of the spot in use). For each one it finds its best three daylight hours, with the same scoring as The best hours, and the best window wins; on a tie, the nearest spot. With the same swell, what usually decides is which way each beach faces and how the wind hits it.
- **Always fresh data:** at the alert's time the watch asks for the forecast right then (the models' latest run), with the phone nearby or over Wi-Fi or LTE. Without a connection it tries again every 15 minutes for two hours; if it still can't, there is no alert that day. It never alerts with a saved forecast.
- **The icon's star** tells how the day looks: **green** if it's good (3 stars or more), **yellow** if it's fair, **red** if it's weak (no window is worth it: the alert says "Weak day" and shows the least bad one).
- **Open the alert** to see the Today page.
- It alerts once a day. If the watch is in sleep mode or Do Not Disturb at that time, it may not vibrate (it depends on how you have it set up).
- **Turning it on:** on the watch, Settings (hold **UP**) → **Morning alert**, and the time in **Alert time** (04:00 to 09:00, every half hour; each START moves to the next one). Also from the Connect IQ app on the phone, in Surf Dawn Forecast's settings; if you turn it on there, open Forecast once on the watch so it gets scheduled. It comes off.
- **When updating from the store,** Garmin Connect may ask you to accept Forecast's new permissions: background (to fetch the forecast at that time with the app closed) and notifications.

The Today page

It opens when you open the alert, or from Settings → **Best spot today**. After 19:00 it shows tomorrow.

- **The bezel is the day, from 05 to 20,** one segment per hour: green for the hours that look good, dark green for fair ones and dimmed for the ones that don't work. A halo marks the best window and the white dot, the time now.
- **Inside:** the best spot, its stars (the forecast's; the ones from your history are in Session), the best window in green, the waves with their period and the wind with where it blows from, in the usual colours: green head-on or offshore, yellow angled or cross-shore, red shadowed or onshore. At the bottom, the second and third spots.
- **A weak day** says "WEAK DAY" in red, with empty red stars and the least bad window.
- **Spots nearby:** with **UP** or **DOWN**, the spots compared from best to worst, each with its window (or "weak"), a bar showing how good it is and its stars (red if it's weak).
- **START or a tap** (or picking one in Spots nearby) opens that spot's forecast, without changing the spot you picked or the face's. **BACK** goes back to the forecast.
- If the alert already downloaded the day's data less than an hour ago, it shows it without asking; otherwise it asks for it (without a connection, START retries).

Buttons

Button	What it does
UP, DOWN or a tap	Spot picker
Swipe sideways	Change day
START	Request the data again
UP held (MENU)	Settings
BACK	Exit (returns to the face)

Forecast settings

Setting	Values (factory default in bold)
Spot	opens the picker
Best spot today	opens the Today page
Morning alert	Yes or **No** (The morning alert)
Alert time	04:00 to 09:00, every half hour (**06:00**)
Waves	**Total** (the whole sea) or Swell (only the groundswell)
Height	**Metres** or Feet
Wind	**km/h** or Knots
Background	**Sea**, Night, Graphite or Forest (each START moves to the next one)

You have to set them here too: Garmin doesn't let one app read another's settings. The factory defaults are the same in all three. The alert and its time can also be changed from the Connect IQ app on the phone.

No connection

If it can't download the data, it shows the latest data it downloaded for this spot: at the top, in place of the date, it says since when, in orange, **"AS OF 07:10"** if it's from today, **"YESTERDAY 07:10"** or **"AS OF SAT 07:10"** if it's older. Forecasts cover three days, so yesterday's still has today and tomorrow, and one from the day before yesterday only has today (the dots at the bottom show how many days are left). As soon as the connection returns they update on their own when you open the app, or with **START**. If a retry fails, it goes back to showing the saved ones.

If there's no data for this spot that reaches today (the latest is three or more days old, or is from another spot), it shows "NO CONNECTION" in red. Check that your phone is nearby with Garmin Connect open and press **START** to retry.

Besides the spot in use, each app keeps the last forecast of the **two spots you used before**: when you go back to one it shows up right away (with since when, as above) while the new one downloads, also without a connection. They're the first thing deleted if the watch needs room.

Each app keeps its own forecast: what Forecast downloaded isn't seen by Session, and the other way around (Garmin doesn't let one app read another's data). To have it at the beach without your phone, open the app you're going to use before you leave.

Where each number comes from and why those models: Data and models.

Surf Dawn Session

Session records your surf session and saves it to Garmin Connect as a **Surfing** activity. It counts your waves on its own, tells you which way to go in the water and, when you get out, shows you a summary. With the stars you give each session, it learns which conditions you like at each spot.



To open it: press and hold the waves subdial on the Surf Dawn face, or find it in the watch's app list ("Surf Dawn Session"). It opens on the **Home** screen.

Before you go in: the Home screen

Home shows you how the sea is right now at your spot, and whether the watch is ready to record.



Where	What it is	What it's for
Top centre	The time	
Orange bezel (top left)	The waves: height, period and an arrow pointing **where they're heading**. The bezel is longer the bigger they are. The arrow is green if they come in head-on to the beach, yellow if they come in diagonally and red if they hit side-on or from behind	Know if there are waves and whether they reach your beach well
Purple bezel (top right)	The wind: speed, where it blows from and an arrow showing where it's going. Green = offshore (from the land), yellow = cross-shore, red = onshore (from the sea)	Know if the sea is clean or choppy

Where	What it is	What it's for
Middle curve	The day's tide, from 5 to 20 h: high tides (turquoise) and low tides (sand) with their time, the next one with its height, a white dot at the current time and the sunset	Pick the right moment to go in
Large name	The spot	
Stars	How many stars you gave to similar conditions at this spot, and in how many sessions ("1 session"). They don't appear if you turned them off or there's nothing to calculate them from yet. See Stars	Know if it's usually good for you right now
Bottom-left bezel	The GPS: the satellite and the bezel in **red** (no signal), **yellow** (searching) or **green** (ready to record)	Wait for green before going in
Bottom-right bezel	The water: degrees and a colour by temperature (blue cold, turquoise, green, yellow, orange warm). It comes from the forecast	
Bottom	The recommended wetsuit (thickness and icons for boots, gloves and hood) and the board you're carrying	Know what to wear

No connection: if it can't download the forecast, it shows the last one it downloaded for this spot (in this app: open it before you head out), with the waves and wind **in grey**, even if it's from yesterday or the day before (it covers three days). The conditions (UP) say since when: "AS OF 07:10", "YESTERDAY 07:10" or "AS OF SAT 07:10". Without a forecast for the spot that reaches today, waves and wind stay as "–" and the conditions say "NO CONNECTION". The session records either way.

Buttons on Home

Button	What it does
START	Starts the session
UP	Shows or hides the day's conditions (the same page as Forecast, with your stars over the best hours). There, swiping sideways changes the day, and START closes them or, if they say "NO CONNECTION", downloads the forecast again
DOWN	Opens the menu: the routines and Settings
Hold UP (MENU)	Settings
BACK	Closes the app (or closes the conditions, if they're open)

The spot. Session picks the nearest spot on its own. With the first good GPS fix it switches to the one nearest to where you really are. If you picked one by hand (Settings → Spot), it keeps it as long as you're within 5 km of it. The picker is the same as in Forecast, with the wave over sand, rocks or reef according to the break type. Each app keeps its own spot: choosing one here doesn't change it on the face or in Forecast.

The recommended wetsuit

At the bottom you'll see the wetsuit for today's water: the thickness as text and, when needed, icons for **boots**, **gloves** and **hood**. The watch doesn't just look at the water:

- With wind of 20 to 29 km/h, it counts the water as one degree colder; with 30 km/h or more (strong wind), two degrees.
- If the air is more than 4 °C colder than the water, one more degree.

Water (already adjusted for wind and air)	Wetsuit	Boots	Gloves	Hood
24 °C or more	rash guard			
21 to 24 °C	shorty			
17 to 21 °C	3/2			
15 to 17 °C	4/3			with strong wind
13 to 15 °C	4/3	✓		with strong wind
10 to 13 °C	5/4	✓	✓	✓
less than 10 °C	6/5	✓	✓	✓

It's a guide: each wetsuit brand shifts the cut-offs by a degree or two.

Routines: Pre-surf, Post-surf and CO2 table

DOWN on Home opens the menu: three routines and **Settings** (the same as UP held). UP and DOWN choose, START starts and BACK goes back to Home. They're meant for home: Pre-surf before you go, Post-surf when you're back. Each one is saved in Garmin Connect as an activity: Pre-surf and Post-surf as **Flexibility training** and the CO2 table as **Breathwork**, with a lap per exercise, the time and the heart rate. If you leave with BACK before the first minute it isn't saved; after that, what you did is. A phone notification over the screen doesn't cut it: it stays one activity.







Routine	How long	What's in it
Pre-surf	4 min	Cat Camel ×6, Bird Dog ×6, Lunge with twist ×3 per side, Arm circles 0:30, Heel raises ×15, Pop-up ×5
Post-surf	10 min	Cat Camel ×6 in 30 s, Doorway chest, Kneeling lat stretch, Open book, Hip flexor, Standing quad, Figure-4 glute, Hamstrings with band and Wall calf stretch (1 min each, 45 s the lats), Neck ×10 in 30 s and 4-6 breathing 1:00
CO2 table	14 min	2 min of easy nasal breathing, 6 holds of 30 s with 1:30 of breathing between them, and 1 min of 4-6 breathing

Everything looks after your back: no loaded bending, a neutral spine, and the movement comes from the hips and the upper back. The lower back isn't stretched (no knees to the chest): after paddling it moves with the Cat Camel, gently and in a comfortable range.

Each exercise is one screen: a figure doing the movement, back and forth (orange is what moves; in the arm circles it goes round, and halfway it turns the other way), the name, the reps or the time, and a cue at the bottom. The bezel has a segment per exercise: teal the ones done and orange the current one, filling with its time.

- **With reps** ("6 times", "3 per side"): when you're done, START goes to the next one.
- **With time** ("1:00"): it waits with "START starts", so you can get in position. With START the countdown runs ("0:18 / 1:00"), it moves on by itself with a vibration at the end, and START skips it. The ones done per side vibrate twice halfway and say "side 2" (arm circles say "now backwards").
- **At a pace** (Post-surf's Cat Camel and neck): they also wait for START and then move on by themselves. The figure sets the pace, once per back and forth, and the number says which one you're on ("4 / 10").
- **4-6 breathing:** a ring with the 10 s cycle, 4 to breathe in (bright teal) and 6 to breathe out, and a dot going round it. In the middle, the nose while breathing in and the wind while breathing out (the wind crossed out is holding, in the CO2 table). At the bottom it says "breathe in through the nose" or "breathe out slowly", and the watch vibrates at each change: one long vibration to breathe in, two short ones to breathe out.
- **The screen:** it turns on at each change (exercise, side, breathing in or out) and every 10 seconds. AMOLED watches don't let an app keep it on for more than a minute in a row, so it may still go dark for a moment.
- BACK goes back to Home at any time.

At the end **Done** shows the exercises and the time ("IN GARMIN CONNECT": it's already saved), and after 4 seconds it goes back to Home by itself.



CO2 table: it trains bearing the CO2, what you feel in a long hold-down. The crossed-out wind is **Hold**, with an orange ring filling with the seconds; between holds, the nose is **Breathe**. At the end it asks "How did it go?": **Easy** makes the next one 5 s longer to hold (up to 1 min) and then 10 s shorter to rest (down to 1 min); **Hard** gives the rest back first and then shortens the hold (not under 20 s); **Just right** keeps it the same. Each option says what the next one will be.

> **Always on dry land**, seated or lying down, never in the water or driving. Before a hold, a normal breath in: hyperventilating is what causes blackouts. Stop at the first strong urge to breathe.

Starting

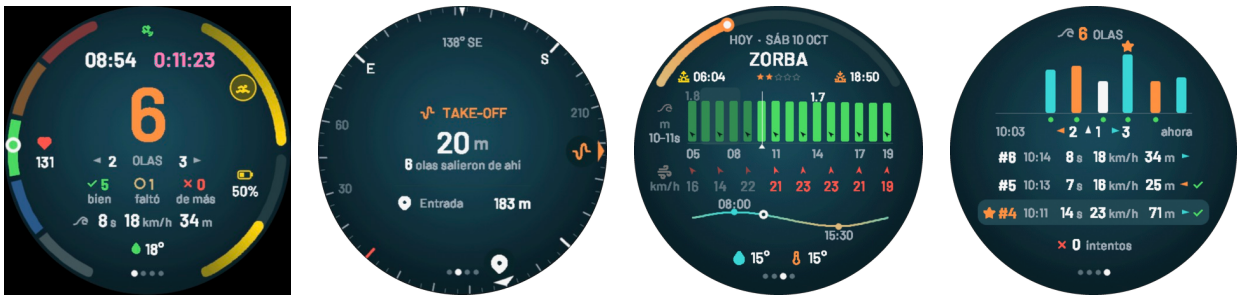
1. Wait for the satellite to turn **green**.
2. Press **START**. The watch vibrates and moves to the Session page.

If there's no GPS yet, START shows the "No GPS" notice. If you still want to start, press **START again**: it records without a position until the GPS shows up.

Tips: - On the watch, set **Display → During activity → Always on**, so you can see it without touching anything. - On a beach that isn't in any zone, press START at the shoreline: with the first metres you paddle, the watch learns which way the beach lies. At spots inside the zones it already knows. - The first minute of the session never counts as a wave.

In the water: the four pages

DOWN goes to the next page and UP to the previous one: **Session → Take-off → Forecast → Waves** and back to Session. If you stay on Forecast or Waves, after 30 seconds it returns to Session on its own. Take-off doesn't return on its own: it stays while you follow the arrow.



Session page

- **Very top, between the two bezels:** the GPS right now: the satellite in **green** (there's a position), **yellow** (weak position) or **red** (no position coming through, for example with your wrist under the water).

- **Top:** the time and, in pink, the session time (pauses don't count).
- **Middle, large:** the waves you've caught so far, with how many went left (◄) and right (►).
- **In Calibrate mode:** your marks: ✓ **good** (you marked it and the watch counted it), ○ **missed** (you marked it and the watch didn't see it), ✗ **extra** (the watch counted one you didn't mark). See Marking waves.
- **The last wave:** how many seconds it lasted, the top speed and the metres.
- **The water:** the temperature the watch measures. It shows up after 8 minutes of session, once the sensor has stopped feeling the heat from your wrist.
- **Left bezel:** your heart rate. The zone you're in lights up (grey, blue, green, orange, red) and next to it go the heart and the beats per minute. With no pulse, or below zone 1, all five stay off.
- **Right bezel, top:** what you're doing, with a colour and an icon: clock = waiting, swimmer = paddling, wave = riding, diver = duck diving, pause = session paused.
- **Right bezel, bottom:** the battery (green, yellow or red).

Take-off page: the compass

It takes you back to the place where your waves break.

- **The ring** is a compass that turns with the watch (the N is in red). At the top it says which way the watch is pointing, for example "138° SE". On watches without a compass (which ones) the ring turns with the direction you're moving; when you're still, after 10 seconds it settles with north at the top and the top just says "N".
- **Two markers on the edge**, each one where its place is: the **orange wave** is the **take-off zone** (the place where you caught today's waves) and the **white pin** is the **entry** (where you started the session). Each marker has a notch that points outward.
- **In the middle**, with the same orange wave: how many metres you are from the take-off and how many waves started there. **At the bottom**, with the same pin: the metres to the entry.
- **Before your first wave**, if you've already surfed that spot, the wave marks the zone from an earlier session (in sand colour): the last one at the same tide (rising or falling), and the bottom says "from your sessions at this tide"; if you've never gone at this tide, the one from your last session there, and it says "from your last session here". If you've never surfed that spot, it says "shows up with the first wave".
- If a marker ends up right behind you, it shifts a little along the edge so it doesn't cover the page dots.
- **The current**, when the watch has measured it: a turquoise arc on the edge toward where it's carrying you, and the row "Current 0.3 m/s". See Current and drift.
- **The swell period** you feel sitting on the board (the stopwatch), next to the one the forecast gives: "Period 11 s · fcst 10".

Distances are always in metres and the current in metres per second.

Forecast page

The conditions: the spot's forecast for the day, as in Forecast, with **your stars** over the three best hours. You can't change day while recording. If it couldn't be downloaded, it shows the last one downloaded with since when at the top ("AS OF 07:10", "YESTERDAY 07:10"), or says "NO CONNECTION" if there's none for the spot that reaches today: while recording, START opens the menu, so to retry, end the session and press START on the conditions in Home.

Waves page

- **The timeline** of the session: one bar per wave, taller the longer it lasted. **Orange** = left, **turquoise** = right, **white** = straight. The **star** marks the best wave of the day, and a green dot, the ones you marked.
- **How many** went left, straight and right.
- **The last three waves:** time, seconds, top speed, metres and which way. The best one is highlighted.
- **Attempts:** the waves you paddled for and didn't catch.

The **best wave** is the one that adds up the most between how long it lasted, how fast it was and the manoeuvres you did.

Buttons in the water

Button	What it does
START	Opens the menu without pausing the session
BACK once	Nothing. That way your wetsuit cuff doesn't kick you out of the app or mark waves by accident
BACK twice in a row	Marks a wave (in Calibrate mode). The watch vibrates twice
UP / DOWN	Previous / next page
Hold UP (MENU)	Settings (the ones you can change in the water)
Hold LIGHT	The watch's controls menu (light, inReach messages, SOS). The session keeps recording
Touchscreen	Does nothing with the session open (recording, paused and on the summary, also in the START menu, Rate and Tags), unless you turn on "Touch in session" in Settings. Settings and the Discard confirmation are watch screens and respond to touch

Vibrations: one at the start, a short one every time the watch counts a wave (when it ends, not when you stand up), two short ones when you mark, a long one with each alert and two when saving.

The START menu



START opens the menu without pausing. UP and DOWN choose, START confirms and BACK closes it. If you don't touch anything, it closes on its own after 20 seconds ("closes in 15 s").

Option	What it does
Continue	Goes back to the session (if it was paused, resumes it)
Mark wave	Same as BACK twice (only if marking is on)
Pause / Resume	Pauses the time and the detection, or turns them back on
Waves	The list of all your waves, newest to oldest: seconds, top speed, metres, which way, **strokes** to catch it and **manoeuvres**; the selected wave (UP and DOWN) also shows its **pop-up** and whether it was **frontside** or **backside**. The best one has a star
Save	Ends the session and shows the summary
Discard	Deletes the session (asks for confirmation)
Mode	Switches from Calibrate to Automatic and back

To finish: START, UP three times to **Save**, and START.

How waves are counted

The watch uses the GPS, accelerometer and gyroscope in your wrist:

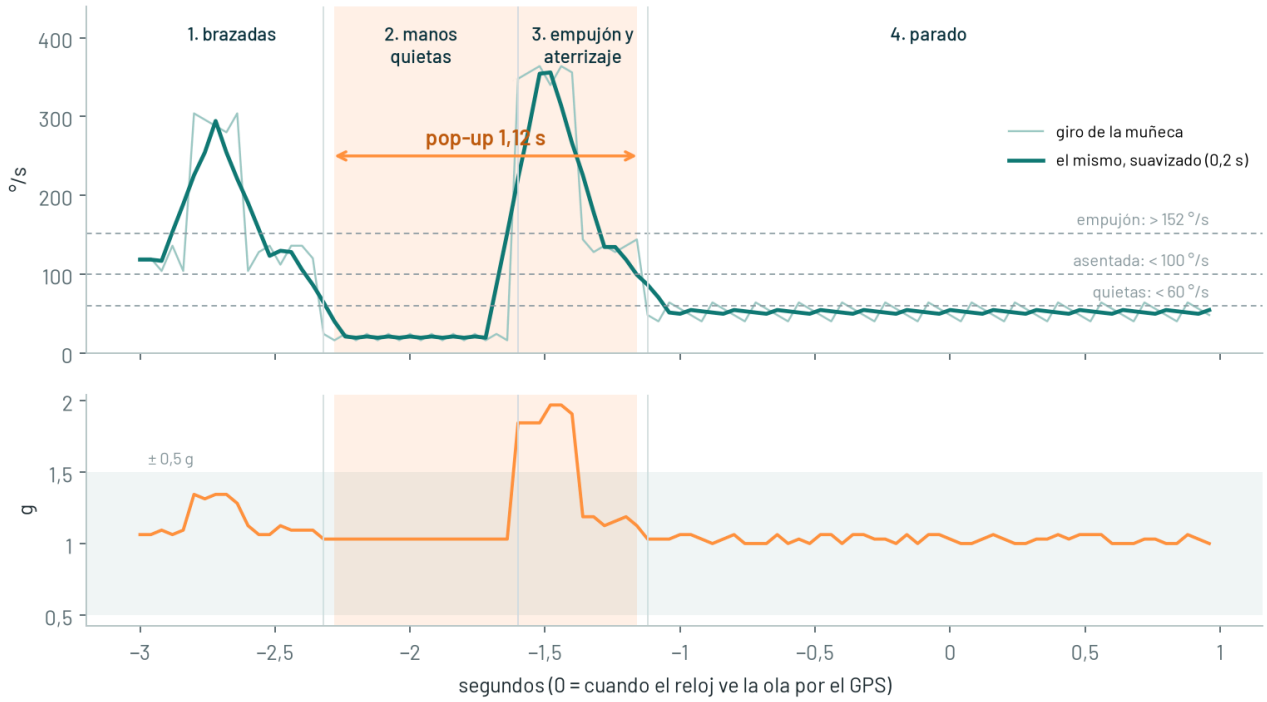
- **A wave** starts when you head toward the beach at more than 7 km/h for a couple of seconds (or with the drop and jolt typical of a take-off on a wave with a wall). It counts if it lasts at least 3 seconds, reaches 10 km/h and covers at least 10 metres. Between one wave and the next, 15 seconds have to pass. These numbers are those of the **Wall** detection, the factory default; in Settings you can choose another.
- **An attempt** is a strong paddle toward a wave that doesn't end up as a wave. They're counted separately.
- **Caught** = waves out of waves plus attempts. If you paddled for 9 and caught 6, it's 6/9 (66 %).
- **Left or right** is seen from the board, facing the beach: if the wave took you toward your right it's a right. If you went straight to the shore, it's "straight".
- **Frontside or backside** comes from the direction and your stance (Settings → **Stance**, Goofy by default): facing the wave (frontside) are the lefts for a goofy and the rights for a regular; the others are backside. A "straight" wave is neither.
- **Manoeuvres:** within a wave, each sharp turn of the wrist (measured by the gyroscope) or each change of direction of more than 60° that the GPS sees counts as a manoeuvre, at most one every 3 seconds. If you turn off the gyroscope in Settings, only changes of direction count.
- **Strokes:** the watch picks up the rhythm of your paddling on your wrist. For each wave it counts the strokes in the 15 seconds before (how much you paddled to catch it). It also adds up those of the whole session and the metres paddled.
- **Pop-up:** the time it takes you to stand up, measured with the wrist. See The pop-up.

Strokes, manoeuvres and pop-up aren't shown while you surf: they're in the wave list in the menu (the pop-up, on the selected wave), strokes and manoeuvres also in the rating, and everything in Garmin Connect.

The pop-up

It's the time it takes you to stand up. The watch measures it with the gyroscope and accelerometer in your wrist, 25 times per second, from the end of your last stroke until you're standing:

1. **Strokes:** while you paddle, the wrist turns hard on every stroke and is never still, because the arm comes back through the air.
2. **Still hands:** you put your hands on the board and push. The wrist moves with the board and barely rotates. **The pop-up starts here.**
3. **Push and landing:** the hands leave the board, the arms go up (the wrist rotates fast) and the accelerometer marks the push.
4. **Standing:** the wrist settles. **It ends here.**



The chart is a take-off built from the timings of a study of recreational surfers (below). Top, the wrist rotation; bottom, the acceleration. The orange band is what the watch measures.

You see it in the wave list (START → **Waves**): the wave selected with UP and DOWN shows its pop-up. It also goes in each wave's lap in Garmin Connect. If the wrist doesn't show that sequence, it says "—" (0 in Connect). That happens when your hands never paused, when you weren't paddling or with the gyroscope off.



It's a first version. The rule was built before there were real take-offs to tune it on, so it can be off. In Calibrate mode each session saves the raw accelerometer and gyroscope, and these sessions will be used to tune the thresholds (how it works).

For reference, what the studies say (all on land; there are no published numbers in the water, nor for beginners):

Who	From → to	Time
Competitive surfers	the chest lifts off → the front foot lands	0.56 s (men), 0.68 s (women)
Experienced recreational surfers	hands touch the board → become stable (what the watch measures)	1.2 s
Intermediates around 25 years old	same	1.1 s
Intermediates around 50 years old	same	2.0 s

Current and drift

When you're sitting there waiting, the watch looks at how much you move without paddling. After about 2 minutes of staying still (and half a minute after your last wave), it calculates the **current**: how fast it carries you and in which direction. It shows up on Take-off and stays until it measures again. If you paddle or catch a wave, it starts measuring again.

While you wait, it also measures the **swell period** from the bobbing of the board. It's experimental: it's good for comparing with the forecast.

If while you wait the current takes you **150 metres** away from your take-off zone, it warns you with a long vibration (see Alerts).

Alerts



Each alert vibrates long and shows a banner that says "the session keeps recording". It closes on its own after 8 seconds, or sooner with BACK or START. They're turned on and off in **Settings → Alerts**.

Alert	When	Default
Sunset	30 minutes before sunset	on
Tide	When high or low tide arrives (at the time it falls)	on

Alert	When	Default
Drift	If, while waiting, you drift 150 m from the take-off zone	on
Every 30 min	Every half hour of session, with the waves you have so far	off

The sunset and the tide come from the forecast, which the watch downloads when you open the app with your phone nearby.

Marking waves: Calibrate and Automatic mode

Calibrate is the default. When each real wave ends, press **BACK twice**. The watch compares your marks with what it counted and shows you good / missed / extra. It also saves a second-by-second log, with the raw accelerometer and gyroscope (25 readings per second), to improve detection and to learn to measure the pop-up with your waves. That log makes the activity bigger (about 0.6 MB more per hour), so it takes a little longer to transfer to Garmin Connect.

When the counts add up for you, switch to **Automatic** (Settings → Mode, or in the START menu): no marks, no log and a cleaner screen. If you still want to mark in Automatic, turn on "Mark with BACK"; those marks stay only in Garmin Connect.

A mark counts for a wave if you make it from a little before you stand up until 45 seconds after it ends, as long as you haven't caught another one. Easiest: mark as soon as the wave ends, or when you paddle back out.

Ending the session

With START → Save → START you get to the summary, the rating and the tags.









Summary (two pages, UP and DOWN switch between them): - The time **in the water** and how it was split: riding (turquoise), paddling (orange) and waiting (grey). - **Waves** (in Calibrate, also how many you marked; in Automatic, how many frontside and backside according to your stance; the lefts and rights are on the Waves page), **top speed**, **the best** (the one with the star on the Waves page: seconds and metres) and **caught** (waves out of attempts). - On the second page: paddling (kilometres and strokes), water, average and max heart rate, and what the swell, wind and tide were like at the start. - START saves and goes on to Rate. BACK goes back to the menu with the session paused: Resume keeps recording. At the bottom it says: "▲▼ more · START saves / BACK back to the menu".

Rate ("How was it?"): starts at 3 stars. UP adds one, DOWN takes one away (from BAD to EPIC). The bezel shows an automatic score from 0 to 10 and at the bottom go two figures from your session, for example "6 strokes on average to catch" or "100 % of attempts caught". START goes on to the tags. BACK saves without stars. At the bottom it says: "▲▼ stars · START next / BACK saves unrated".

Tags: Size, Wind, Crowd, Current, Section and Tide. They help you remember what was special about that session. UP and DOWN move between the tags (the one with the white border), START turns one on or off (when on it turns **green**) and BACK saves. At the bottom it says: "▲▼ choose · START marks / BACK saves".

When saving, "Saved · IN GARMIN CONNECT" appears, the watch vibrates twice and goes back to Home, ready for another session. If you turn off "Rate when saving", START on the summary saves straight away.

If the watch closes the app while you're recording, the session is saved anyway, without stars.

Stars

Every time you give stars to a session, the watch saves what the conditions were like at the start: waves, period, where the swell was coming from, wind, tide and time. Then, for any hour of the forecast, it looks for your sessions **at that same spot** with similar conditions and shows you the average of the stars you gave them. The most similar ones weigh more.

- With few similar sessions (fewer than 3), it blends your stars with an estimate from the forecast (size, period, wind and beach orientation). With none, you see only the estimate.
- "1 session", "4 sessions" next to the stars says how many similar sessions it's based on.
- Stars appear on **Home** (for right now), on the **Forecast** page (over the best hours of the day) and in the **Best window** complication (see below).
- In Settings → Stars you choose "History + estimate" (default), "History only" or "No". With "No" they don't appear anywhere; with "History only", only when you've rated similar sessions at that spot.

The more sessions you rate, the better it knows you.

History and records



Settings → History shows: - The **records of the spot** you're at: most waves in a session, the longest wave and your best rating. - **Your best conditions** at that spot: what the waves, wind and tide were like in the session you gave the most stars to. - **Your sessions**, newest to oldest (from all spots): date, waves, the longest and stars. UP and DOWN scroll through the list.

To see another spot's records, change the spot first (Settings → Spot). The watch keeps up to 60 sessions per spot and 200 in total; when it hits the limit the oldest ones are deleted.

Boards

The board you're carrying appears above START and is saved in each session (also in Garmin Connect, in the **Board** field).

- **Choosing it:** Settings → Board. Each one shows its length, litres and how many sessions it has. It stays selected for the next sessions. The board is fixed when you press START: you can't change it in the water.
- **Adding your own:** on the phone, Connect IQ app → Surf Dawn Session → Settings → Boards (up to 12, with model, length and litres). The app doesn't come with boards: until you add them, sessions are recorded without a board. Until you pick one, the first on your list is used.
- **None:** records the session without a board.
- **Remove a board:** takes it off the watch's list (asks for confirmation). To get it back, add it again from the phone.

Settings

Hold **UP** (MENU) on Home or while recording, or choose **Settings** in the DOWN menu. START changes the value or opens the option, BACK closes.



Setting	Options (defaults in bold)	What it's for
Board	your boards, None, Remove a board	See Boards
Stance	**Goofy**, Regular	Which foot you surf with forward: with that each wave comes out as frontside or backside
Watch on	**left**, right	Which wrist you wear the watch on. It's saved with each wave in Garmin Connect to tune the pop-up
Spot	the spot picker	Choose the spot by hand
Detection	**Wall**, Whitewater, Performance, Custom	**Whitewater** for small, slow waves or whitewater; **Wall**, the everyday one; **Performance** for big, fast waves (asks for more speed and distance)

Setting	Options (defaults in bold)	What it's for
Custom	Take-off (**7** km/h), Minimum peak (**10** km/h), Minimum distance (**10** m), Minimum duration (**3** s), Between waves (**15** s), Towards the shore (**yes**)	Your own thresholds. Changing one switches detection to Custom. "Towards the shore: no" also counts waves that don't head toward the beach
Mode	**Calibrate**, Automatic	See Marking waves
Mark with BACK	yes, **no**	Mark waves in Automatic too
Vibrate on detection	**yes**, no	One vibration for each wave counted
Water temperature	**Sensor**, Forecast, Both	Which water to show and save: the one the watch measures (from 8 minutes in), the forecast's, or the forecast until the sensor is ready. On watches without a thermometer, Sensor shows the forecast's
GPS	**Multi-band**, SatIQ, GPS only	Multi-band is the most accurate; SatIQ uses less battery. Only the ones your watch takes show up: the fēnix 9 Pro takes SatIQ only. On watches without multi-band, Multi-band uses all systems on one band
Gyroscope	**yes**, no	Counts manoeuvres with wrist turns; off, it uses less battery. On watches without a gyroscope nothing changes
1 Hz log	yes, **no**	Save the second-by-second log (with the raw accelerometer and gyroscope) in Automatic too
Lap per wave	**yes**, no	One lap per wave in Garmin Connect. Off, the activity is a single lap, with no wave data
Alerts	Sunset, Tide, Drift, Every 30 min	See Alerts
History		See History and records
Rate when saving	**yes**, no	Ask for stars and tags when saving
Stars	**History + estimate**, History only, No	See Stars
Waves	**Total**, Swell	Total = the whole sea (like the "Wave" row on Windguru); Swell = just the ground swell
Height	**Metres**, Feet	
Wind	**km/h**, Knots	Also the speed of the waves

Setting	Options (defaults in bold)	What it's for
Touch in session	yes, **no**	Use the touchscreen with the session open
Background	**Sea**, Night, Graphite, Forest	The background colour, as on the face

While recording, Board, Stance, Watch on, Spot, Detection, Custom, GPS, Gyroscope, 1 Hz log, Lap per wave and History don't appear: you choose them before starting. These settings belong to Session only: Garmin doesn't let one app read another's, so units and background are chosen in each app.

What ends up in Garmin Connect

The session shows up as **Surfing**, with your route and your heart rate, and also:

- **One lap per wave** (if "Lap per wave" is on), with how long the wave lasted, the metres, the side, whether you marked it, the max and average speed, the strokes to catch it, the manoeuvres, the pop-up (0 when it couldn't be measured) and whether it was frontside (1) or backside (-1). Off, you get a single lap with that data empty; the waves are still counted in the session data.
- **Session data:** waves, attempts, the longest, minutes riding and paddling, strokes, distance paddled, water, what the watch measured (swell period and current), the conditions at the start, spot, board, stars, tags and score.
- **In Calibrate mode**, two more charts: what you were doing (waiting, paddling, riding, duck dive) and the paddling.

The routines are separate activities: Pre-surf and Post-surf as Flexibility training and the CO2 table as Breathwork, with a lap per exercise.

Garmin Connect shows this data under its own names when the app is installed from the Connect IQ store. If you installed it by cable, it's still inside the activity file.

Session on other watch faces

Session publishes two complications that any compatible face can show: - **Surf Dawn Session**: your last session ("8 waves"). From the Surf Dawn face you open it by pressing and holding the waves subdial. - **Surf Dawn Best window**: the two hours of the day with the most stars for you ("14-16 h · 4/5"). When you open Session from there, it starts on the day's conditions.

Both are calculated every time you save a session: the best window is the one for the day you saved the last one.

How waves are detected, in detail, and how it was tested: Session design.

Install, watches and FAQ

Install

- **From the Connect IQ Store:** the **Surf Dawn** face and **Surf Dawn Forecast** are free: search for them in the Connect IQ app on your phone, install them and they update themselves. You change their phone settings from there.
- **Session** is still a private beta on the developer's account: it only shows up in the Connect IQ app of that account, and its phone settings (for example, the boards) are changed from there. The face and Forecast work without it.
- **Session on someone else's watch:** Garmin doesn't allow sharing a beta, so it goes over a cable. You need the app files (`.prg`) for that watch model. With the watch set to **Settings → System → USB Mode → MTP**, you copy them to the `GARMIN/Apps` folder (on Windows the watch shows up as a device; on a Mac you need an app such as OpenMTP). When you unplug it they are installed. This way there are no settings from the phone, and Garmin Connect saves Session's data without its names.

Supported watches

All three apps run on Garmin **round watches with 5 buttons** and Connect IQ 4.2 or later: 48 models.

Display	Watches
AMOLED	fēnix 9 Pro (43, 47 and 51 mm), fēnix 9 (43, 47 and 51 mm), fēnix 8 (43, 47 and 51 mm), fēnix 8 Pro (47 and 51 mm, also MicroLED), tactix 8, quatix 8 and 8 Pro, fēnix E, epix Pro (Gen 2) in 42, 47 and 51 mm, epix (Gen 2), quatix 7 Pro and Sapphire, tactix 7 AMOLED, MARQ (Gen 2) and MARQ Aviator, D2 Mach 1, Mach 1 Pro, Mach 2 and Mach 2 Pro, Forerunner 965, 970, 570 (42 and 47 mm), 265, 170, 165 and 70 (and their Music versions), Descent Mk3 and Mk3i 43 mm, Descent Mk3i 51 mm, Descent G2, Instinct 3 AMOLED (45 and 50 mm) and Instinct Crossover AMOLED
MIP (transflective display)	fēnix 7, 7 Pro and 7 Pro Solar, fēnix 7X, 7X Pro and 7X Pro Solar, tactix 7, quatix 7 and 7X Solar, Enduro 2 and 3, fēnix 8 Solar (47 and 51 mm), tactix 8 Solar, fēnix 9 Pro Solar (47 and 51 mm), Forerunner 955 (and Solar), 255 and 255 Music

Differences you will notice:

- **MIP:** flat colours and the face has no second hand. The display never turns off, so at rest you still see the full face.
- **Instinct Crossover AMOLED:** it has real hands, so the face doesn't draw its own and has no digital-time option.

- **No touchscreen** (Forerunner 255 and 255 Music, Instinct 3 AMOLED, Instinct Crossover AMOLED): the face gestures don't exist (open Forecast and Session from the app list) and Forecast shows a single day.

Watches with 2 or 3 buttons (Venu, vívoactive, Approach, D2 Air), monochrome-display Instincts, smaller displays (fēnix 7S, Forerunner 255s and 265s) and watches older than Connect IQ 4.2 (fēnix 6, first-generation MARQ, Forerunner 945) are not supported. The table with every model is in the README.

What each watch measures

All of them have GPS, an accelerometer and a heart-rate sensor: that is enough to count waves, paddling and distance on any of them. The other sensors vary by model (Garmin spec sheets, October 2026).

- **Price** compares Garmin's list price when the model came out, not what it costs in any particular country: **$** up to US$350, **$$** up to US$650, **$$$** up to US$1,000, **$$$$** up to US$1,500 and **$$$$$** above that. The table runs from cheapest to most expensive; with sales or in another country the number changes, but it is good for comparing.
- **Session** is a star rating: how many of the five sensors Session makes use of the watch has (gyroscope, barometer, thermometer, compass and multi-band GPS). The touchscreen doesn't count, because in the water Session is operated with the buttons.

Watch	Price	Session	Gyroscope	Barometer	Thermometer	Compass	Multi-band GPS	Touch
Forerunner 255 and 255 Music	$	★★★★★	yes	yes	yes	yes	yes	**no**
Forerunner 170 and 170 Music	$	★★★★☆	yes	yes	yes	yes	**no**	yes
Forerunner 165 and 165 Music	$	★★★☆☆	**no**	yes	yes	yes	**no**	yes
Forerunner 70	$	☆☆☆☆☆	**no**	**no**	**no**	**no**	**no**	yes
Forerunner 265	$$	★★★★★	yes	yes	yes	yes	yes	yes
Forerunner 955 and 955 Solar	$$	★★★★★	yes	yes	yes	yes	yes	yes
Forerunner 570 (42 and 47 mm)	$$	★★★★★	yes	yes	yes	yes	yes	yes
Forerunner 965	$$	★★★★★	yes	yes	yes	yes	yes	yes
Instinct 3 AMOLED (45 and 50 mm)	$$	★★★★★	yes	yes	yes	yes	yes	**no**
Instinct Crossover AMOLED	$$	★★★★★	yes	yes	yes	yes	yes	**no**
Descent G2	$$	★★★★☆	yes	yes	yes	yes	**no**	yes

Watch	Price	Session	Gyroscope	Barometer	Thermometer	Compass	Multi-band GPS	Touch
Forerunner 970	$$$	★★★★★	yes	yes	yes	yes	yes	yes
fēnix 7 Pro and 7 Pro Solar	$$$	★★★★★	yes	yes	yes	yes	yes	yes
fēnix 7X Pro and 7X Pro Solar	$$$	★★★★★	yes	yes	yes	yes	yes	yes
fēnix 7 Sapphire Solar	$$$	★★★★★	yes	yes	yes	yes	yes	yes
fēnix 7X Sapphire Solar	$$$	★★★★★	yes	yes	yes	yes	yes	yes
epix Pro (Gen 2) 42, 47 and 51 mm	$$$	★★★★★	yes	yes	yes	yes	yes	yes
epix (Gen 2) Sapphire	$$$	★★★★★	yes	yes	yes	yes	yes	yes
quatix 7 Sapphire	$$$	★★★★★	yes	yes	yes	yes	yes	yes
quatix 7X Solar	$$$	★★★★★	yes	yes	yes	yes	yes	yes
Enduro 3	$$$	★★★★★	yes	yes	yes	yes	yes	yes
fēnix 9 (43, 47 and 51 mm)	$$$	★★★★★	yes	yes	yes	yes	yes	yes
fēnix 8 (43, 47 and 51 mm)	$$$	★★★★★	yes	yes	yes	yes	yes	yes
fēnix 8 Solar (47 and 51 mm)	$$$	★★★★★	yes	yes	yes	yes	yes	yes
fēnix 7 and 7 Solar (without Sapphire), quatix 7	$$$	★★★★☆	yes	yes	yes	yes	**no**	yes
fēnix E	$$$	★★★★☆	yes	yes	yes	yes	**no**	yes
fēnix 7X Solar (without Sapphire)	$$$	★★★★☆	yes	yes	yes	yes	**no**	yes
epix (Gen 2) without Sapphire	$$$	★★★★☆	yes	yes	yes	yes	**no**	yes
fēnix 9 Pro (43, 47 and 51 mm)	$$$$	★★★★★	yes	yes	yes	yes	yes	yes

Watch	Price	Session	Gyroscope	Barometer	Thermometer	Compass	Multi-band GPS	Touch
fēnix 9 Pro Solar (47 and 51 mm)	$$$$	★★★★★	yes	yes	yes	yes	yes	yes
Descent Mk3 and Mk3i 43 mm	$$$$	★★★★★	yes	yes	yes	yes	yes	yes
Enduro 2	$$$$	★★★★★	yes	yes	yes	yes	yes	yes
tactix 7 and tactix 7 AMOLED	$$$$	★★★★★	yes	yes	yes	yes	yes	yes
quatix 7 Pro	$$$$	★★★★★	yes	yes	yes	yes	yes	yes
quatix 8	$$$$	★★★★★	yes	yes	yes	yes	yes	yes
fēnix 8 Pro (47 and 51 mm)	$$$$	★★★★★	yes	yes	yes	yes	yes	yes
tactix 8 and tactix 8 Solar	$$$$	★★★★★	yes	yes	yes	yes	yes	yes
D2 Mach 1 and Mach 1 Pro	$$$$	★★★★★	yes	yes	yes	yes	yes	yes
quatix 8 Pro	$$$$	★★★★★	yes	yes	yes	yes	yes	yes
D2 Mach 2	$$$$	★★★★★	yes	yes	yes	yes	yes	yes
D2 Mach 2 Pro	$$$$$	★★★★★	yes	yes	yes	yes	yes	yes
Descent Mk3i 51 mm	$$$$$	★★★★★	yes	yes	yes	yes	yes	yes
MARQ (Gen 2): Athlete, Adventurer, Captain, Golfer, Carbon, Commander Carbon	$$$$$	★★★★★	yes	yes	yes	yes	yes	yes
fēnix 8 Pro MicroLED	$$$$$	★★★★★	yes	yes	yes	yes	yes	yes
MARQ (Gen 2) Aviator	$$$$$	★★★★★	yes	yes	yes	yes	yes	yes

Zero stars doesn't mean it doesn't work: the Forerunner 70 counts waves, paddling and distance like any other. The face and Forecast work the same on all of them, because they use the GPS and the phone's data, not these sensors.

What changes in Session when the watch lacks a sensor:

If it doesn't have	What you lose	What Session does instead
Gyroscope	Manoeuvres from wrist rotations and the pop-up time	It counts manoeuvres from GPS changes of direction. The pop-up reads "—" and the Gyroscope setting changes nothing
Barometer	Knowing when you duck under a wave: no "duck dive" appears and it can't use it to discard false waves	It counts waves just the same, with the GPS and the accelerometer
Thermometer	The water temperature measured at your wrist	Water temperature → Sensor shows the forecast's value 30 seconds after the start
Compass	The Take-off ring following where the watch points when you are still	While you move, the ring turns with your heading (GPS). Standing still, after 10 seconds it settles with north up and the top just says "N"
Multi-band GPS	Extra accuracy near cliffs, piers or buildings	GPS → Multi-band uses all systems on a single band
Touchscreen	The face's gestures and seeing more than one day in Forecast	Everything is operated with the buttons; Forecast and Session open from the app list

Languages

The apps are in **Spanish** if the watch is set to Spanish, and in **English** if it is set to any other language. The same goes for the names of Session's data in Garmin Connect.

inReach messages

Watches with inReach (such as the fēnix 9 Pro) send and receive messages by satellite. No Connect IQ app can read or send them, but the watch's own access works on any screen, even while recording a session (the session keeps recording):

1. Hold **LIGHT**: the controls menu opens.
2. Choose **Messenger** to read and reply, or **Check-in** to send your "I'm OK" with your location. **SOS** is there too.

To have Messenger first: with the controls menu open, hold **MENU** → Reorder Controls, choose Messenger and move it up. If it doesn't appear, add it with Add Controls. From the face, the inReach glance is in the glance list (DOWN).

With the phone far away or out of signal, the watch uses the satellite. New messages also arrive as a notification during the session, if you have notifications during activities turned on.

If your watch offers Messenger to watch faces, touching and holding the top row of the Surf Dawn face (sunrise and sunset) opens it.

Frequently asked questions

The numbers are grey and it says "4 H AGO". No new data has arrived for more than 3 hours. The phone has no connection or Garmin Connect is closed. If there is also a crossed-out phone, the watch isn't connected to the phone.

It says "LOADING" or "—". It is a new spot and its data hasn't arrived yet. Nearby spots load instantly; a distant one takes a few minutes, because Garmin only allows an update once every 5 minutes.

Total or Swell? Total is the whole sea: the groundswell plus the waves the wind builds. It is what Windguru shows in the "Wave" row and it is the default. Swell is the groundswell only.

Does the tide include the wind? No. It is the astronomical tide from the official table of the nearest port. In a sudestada (a strong southeasterly storm surge) the water rises higher than it says.

The face, Forecast and Session show different spots. The spot you choose in Forecast also carries over to the face. But Session picks its own with the GPS when it starts, and the fixed spot from Customize doesn't carry over to Forecast. Garmin doesn't let one app read what another one saved.

Why do I have to choose units and background in each app? For the same reason: each app stores its own settings. The defaults are the same in all three.

The watch missed a wave, or counted an extra one. Use Calibrate mode and mark your waves with BACK twice: you will see how many it got right. If the waves at your spot are small and slow, try the Whitewater detection; if they are big and fast, Performance (Session settings).

Does it measure the pop-up? Yes, as an estimate: it lasts about a second and the watch samples your wrist 25 times a second, so that is enough. You see it in the wave list (the selected wave) and in each wave's lap in Garmin Connect. It is a first version, built before there were real take-offs to test with, and it can be off: in Calibrate mode each session saves the raw accelerometer and gyroscope data so it can be tuned with your waves (how it is measured).

The water says "—" at the start of the session. The watch's sensor needs 8 minutes to stop feeling the heat of your wrist. If you want a number from the start, choose Water temperature → Both. Watches without a thermometer show the forecast's value after 30 seconds (which ones).

What happens if the app closes while I'm recording? The session is saved anyway, with no stars.

Does it show up in Garmin Connect? Yes: as Surfing, with one lap per wave. Session's own data (waves, strokes, board, stars) appears under its name if the app is installed from the store; otherwise it stays inside the activity file.

How do I use less battery? In Session, GPS SatIQ instead of Multi-band and the gyroscope off (it won't count manoeuvres from your wrist). On the face, the Graphite background.

Is Garmin's own Surf activity still useful? Yes, and it is good for comparing. If you use it, set GPS to "All Systems + Multi-Band" and recording to "Every Second".

Appendix: data and models

Where each number on the face and in the forecast comes from, why those models, and what their limitations are.

Where each number comes from

Element	Data	Source / rule
Sunrise / sunset	Next sunrise and sunset	Open-Meteo forecast (`daily.sunrise/sunset`, today and tomorrow)
Battery	The watch's battery %	Traffic light by level: green > 50 %, yellow 20–50 %, red < 20 %
Stale data / no phone	Below the date (in Digital, below the battery), only when something fails	With data older than 3 h the values show **in grey** and "4 H AGO" appears in red; with no connection to the phone a **crossed-out phone** appears (in that state no updates arrive)
Water	Sea temperature	Open-Meteo marine `sea_surface_temperature` (default model, Météo-France; requested together with the tide at the spot in use and shared with its neighbours)
Outside	Air temperature + weather icon; the **number's colour is the current UV risk** (white with no sun, green low, yellow moderate, orange high, red very high, violet extreme)	Open-Meteo forecast `temperature_2m`, `uv_index`, `weather_code` (WMO code) and `is_day`; if missing, the Garmin's own weather

Element	Data	Source / rule
Waves	Height (m), period (s) and direction of origin; at rest also in the row with the wave icon. **Swell traffic light:** the arrow and the period are green if it comes in head-on to the beach (up to 35° off), yellow if diagonal (up to 75°) and red if from the side or from behind (beach blocked)	Open-Meteo marine, with the wave model of the spot's zone (Models): in Uruguay and Santa Catarina **NOAA GFS Wave** (`models=ncep_gfswave025`, the same one Windguru shows). When the model cell nearest to the spot is land (it returns 0.0 m, as on the east coast of Florianópolis with the 0.25° grid), the spot carries an offshore wave point (columns 6 and 7 of its row in the zone, computed by `build_zones.py` in the data repo) and the waves are requested there. The **total** sea (`wave_*`) by default — it's Windguru's "Wave" row — or only the underlying **swell** (`swell_wave_*`) if chosen in Customize (note: the model's swell partition is unreliable off Uruguay; in zones with ECMWF WAM there is no separate swell and the total is shown)
Wind	km/h, compass point it comes from	Open-Meteo forecast, with the zone's wind model; **GFS** by default (`models=gfs_seamless`, Windguru's "GFS 13 km", which over the continental US blends in 3 km HRRR; it also provides temperature, UV and weather code). Colour according to where the spot's beach faces: **green offshore** (within ±45° of the wind blowing straight off the land), **yellow cross-shore** (up to 110°) and **red onshore**

Element	Data	Source / rule
Tide	24 h curve with high and low tides, time and current height	The table for the **zone's port** nearest to the spot (each zone brings its ports with the radius they cover), hourly heights served month by month from the public repo surf-dawn-data, which updates itself (`shared-bg/TideTable.mc`, shared with the apps). In Uruguay, the **official SOHMA table** (Montevideo up to 89 km: Playa Honda, La Pantalla and the Costa de Oro up to Costa Azul; Punta del Este up to 60 km: from Biarritz to Jaureguiberry and from La Virgen to José Ignacio; La Paloma up to 135 km: all of Rocha up to Chuy, with at most a 10–20 min offset; each spot uses the nearest port that covers it), in metres above each port's zero level. In Santa Catarina, the **official DHN (Marinha do Brasil) table for Imbituba** (up to 100 km: from Praia da Vila to Praia Brava), taken from the annual PDF that Epagri/CIRAM republishes. In the rest of the world, in order: the official predictions that can be republished (**NOAA** in the US, Hawaii, Puerto Rico and the Pacific; **JMA** in Japan; **Marine Institute** in Ireland; **LINZ** in New Zealand) and, where there are none, the harmonic constants from **TICON-3** for the nearest tide gauge on the same coast (`tools/tide/build_tides.py` in the data repo; mean level plus the sum of amplitudes, always positive); where there is no tide gauge nearby (Argentina, Central America, much of Indonesia, the Maldives, Africa…), NASA's global **GOT5.6** model at the central spot of each group of spots within 50 km. That way every zone has tide by port. If the month can't be downloaded, the port's harmonics (`harmonics.json`, ±4 cm in Uruguay). Outside the radii, Open-Meteo's hourly `sea_level_height_msl`. In every case the curve looks for maxima and

Element	Data	Source / rule
		minima, refines them with a parabola and joins them with half-cosines. It is astronomical tide: it does not include wind.

The data is requested in the background every **30 min** (`REFRESH_SECONDS`), and every **15 min between 5 and 10** (`MORNING_*`), the first hours of surfing. If one of the two APIs fails, the last values from that API are kept. Values turn grey after 3 h without new data ("N H AGO", and "N D AGO" from 48 h).

Which spot and which data the face requests

- **Spot:** the one pinned on the watch (Storage `spotPick`, an id `<REGION>/<NAME>`); if there is none, the nearest one within **25 km** (`Spots.RADIUS_KM`) of the last known position, among the watch's zones; if there is none, the one set in `properties.xml` or from the phone (La Plage by default). The position is the last one the watch has (`Geo.lastLocation`, which does not turn the GPS on): that of its last GPS fix or, without a GPS activity, the one the phone passes it through Garmin Connect, only as precise as the phone lets Garmin Connect be (on a fēnix 9 Pro in Punta del Este it was 19 km off, and the spot nearest to that point was Punta del Chileno; opening Session until the GPS turned green corrected it). The phone's weather position is the same point; it only counts if there is none or if it arrived later and more than 50 km away (`STATION_KM`: you went somewhere else). It is checked every 10 min (`SPOT_CHECK_SECONDS`); in development builds, `SpotPicker.logWhere` writes both positions and which one it took to `SURFDAWN.TXT`.
- **Zones:** spots come in zones of up to 60 from the data repo (`spots/`, format in its `docs/formatos.md`). Uruguay's come with the apps (`shared/resources/jsonData/factory_<REGION>.json`, one resource each, `tools/sync_zones.py`); the others, Santa Catarina included, are downloaded: where there is no zone, the face's background process downloads the 10° cell of the position (`spots/g/`) and, on the next event, the zone that covers it and the watch doesn't have (`spots/r/<REGION>.json`, 2–3 KB), instead of that time's weather (`ZoneLoader`, Storage `zneed`; an unanswered request is repeated once and then waits 6 h); Forecast and Session download them themselves, also by hand from the picker. Each app keeps up to 10 zones besides the factory ones (Storage `z:<REGION>`, `zu`); when there is no room, the least-used automatic one is deleted (an automatic zone never evicts one downloaded by hand), and everything that gets written goes through `Store`, which frees space (old automatic zones, saved tides, the saved forecast, any automatic zone and, only for what the user chooses, the hand-downloaded ones) before giving up, so the watch never gets stuck. Session's history is never deleted to make room for anything else: when the history itself doesn't fit, after the old automatic zones and the saved data, its oldest tenth goes. A zone whose version changed in the repo is downloaded again at most once a week, even while you're at its spots; one that the background process had to trim (it didn't fit in its response) is downloaded again, trimmed around the new place, when you move away from the spots it saved.
- **Preloaded spots:** on every update it downloads the data for the spot in use **and for those in its zone within 25 km** (`PREFETCH_KM`, up to 8: the package the background process returns has a size limit, which is why they go as lists by position, see `SurfKeys.mc`; Storage `prefetch` → cache `surfBy`). It is just 3 requests: the table or the hourly tide at the spot in use (it serves that whole coast and is shared with neighbours within 30 km) and, each in a single Open-Meteo multi-location query, the current sea and weather for all of them. A far-away spot chosen by hand waits for the next background request (Garmin allows one every 5 min; the face retries every 6) and in the meantime shows "—" and LOADING.

- **Orientation:** each spot has the direction its beach faces, computed from the OpenStreetMap coastline (`build_zones.py` in the data repo). It decides the swell and wind colours.

Forecast and Session data

Both apps request the data themselves when they open (and Forecast also with START or when you pick a spot), without the 5 min limit of the background process: hourly Open-Meteo with the waves and wind from the zone's models (the waves at the zone's offshore point when needed), air and sun from the wind model, water from the default marine model and tide from the same `TideTable` as the face (months saved in Storage `tideTab`; harmonics if the month can't be downloaded). Periods and temperatures are rounded like Windguru (9.55 s → 10 s). After 19 h the forecast starts on tomorrow.

Models by zone

Each zone brings its own models (`"m"`: waves and wind), chosen in the data repo (`models.csv`):

- **Uruguay, Santa Catarina and Argentina: GFS Wave 0.25° + GFS**, like Windguru, the reference for those who surf there (decided in issue #7; see below).
- **The rest:** by default GFS Wave 0.16° between 15°S and 52.5°N (0.25° outside that) and GFS. Where there are buoys, the data repo compares Open-Meteo's models against them every month (`validate/`: NDBC, CDIP, IMOS/AODN in Australia and Ireland's Marine Institute) and the zone's model changes (its `models.csv`) when another one is clearly less wrong (high or medium confidence). Only models with **peak period** are chosen (the watch shows the period and scores the hours with it): GFS Wave (its `wave_period` is that of the main system) or **ECMWF WAM** (`wave_peak_period`; it gives the total sea, with no separate swell). In the October 2026 validation ECMWF WAM won on the US West Coast, Hawaii, Baja California, the Gulf of Mexico, the US Northeast, British Columbia, South Australia, Northern Ireland and Clare/Kerry: between 5 and 60 % less height error than GFS Wave (20–45 % in most; in Hawaii and Los Angeles, half). For wind, ECMWF IFS won in the US Southeast and GEM (Canada) in Puerto Rico; everywhere else GFS stays. MFWAM and DWD EWAM only provide mean period in Open-Meteo and are not chosen.

Uruguay: why GFS Wave + GFS

On 2026-10-07 Open-Meteo's models were compared with what Windguru shows for Punta del Este (Parada 31), which is the user's reference in Uruguay. The waves in Windguru's "WG" and "GFS 13 km" tables are those of its **GFS-Wave 25 km** tab (NOAA), and its wind is **GFS 13 km**. In Open-Meteo, `ncep_gfswave025` gave practically the same (e.g. 2.06 m / 9.6 s against 2.1 m / 10 s), while the default model (Météo-France, `best_match`) gave 0.3–0.4 m less and 2–3 s less period, and the `best_match` wind was off by up to 5 knots; `gfs_seamless` matched. That is why both apps request those two models. The tide (`sea_level_height_msl`) and the water temperature do not exist in GFS Wave and are still requested from the default model.

Spot positions and orientation: derived from the coastline of © OpenStreetMap contributors (ODbL). Tides outside Uruguay and Brazil: NOAA/NOS/CO-OPS (public domain), JMA (Public Data License 1.0), Marine Institute (CC BY 4.0), LINZ (CC BY 4.0, Crown copyright), TICON-3 (Hart-Davis et al. 2022, CC BY 4.0, GESLA-3 data) and GOT5.6 (R. Ray, NASA GSFC, public domain).

The best hours (Forecast)

Each hour with daylight gets a score from 0 to 1 (`Conditions.quality` in `shared/`):

- **Waves:** 65 % from height (0 at 0.3 m, full at 1.8 m) plus 35 % from period (0 at 5 s, full at 13 s).
- **Wind:** subtracts nothing up to a speed that depends on direction (offshore 35 km/h, cross-shore 12, onshore 8). Above that, the factor drops linearly until at 3 times that limit it is down to 15 % of the score.
- **Swell exposure:** × 1 head-on (up to 35° off), × 0.75 diagonal (up to 75°), × 0.35 blocked. With no known orientation it counts as blocked and onshore.

The band marks the 3 consecutive hours with the best average, all with daylight (from the sunrise hour, or the next one if sunrise is at the half hour or later, until the sunset hour), and only if that average exceeds 0.1. Today it can fall on hours that have already passed.

The stars (Session)

`StarsModel` in `session-app/source/History.mc`. For the conditions of an hour (height, period, swell deviation, wind, deviation from offshore, tide, whether it is rising or falling, time of day), each rated session (1 to 5) **at the same spot** weighs `exp(-z/2)`, where `z` is the sum of the square of each difference divided by its scale: 0.3 m, 2 s, 30°, 8 km/h, 45°, 0.4 m and 2 h. With the tide in the other phase the weight is divided by two. The stars are the weighted average. "N sessions" counts those with a weight above 0.2.

- **History + estimate:** with fewer than 3 similar sessions it adds the forecast's estimate (`quality` × 5) with a weight of 1.5 sessions.
- **History only:** nothing if the total weight doesn't reach 0.05.
- **Best window:** the two consecutive hours between 6 and 20 with the most stars; it is computed when each session is saved.

Recommended wetsuit (Session)

`Conditions.suit`: the water temperature (from the forecast for the current hour), one degree less with wind of 20 to 29 km/h, two with 30 km/h or more, and one less if the air is more than 4 °C below the water. The cutoffs (table in the manual and in the README) come from the guides by Rip Curl, Vissla, evo and Sendy, which agree on the order (4/3, then boots, then gloves and hood) and differ by a degree or two between brands.

Wave detection (Session)

How waves, attempts, manoeuvres and strokes are detected, with their thresholds and tests: Session design (Spanish).

Pop-up (Session)

`PopUpRule` (`session-app/source/PopUp.mc`, twin of `tools/popup.py`): a provisional rule, written before having real take-offs to tune it with. When the detector validates a wave, the watch looks at the wrist from 6 s before to 6 s after the second it started: |a| in 1/32 g and |ω| in 4 °/s, 25 per second, with |ω| smoothed over 0.2 s.

1. **The push:** a stretch with the wrist rotating faster than 152 °/s, with the accelerometer deviating at least 0.5 g, up or down, around it (the arms leave the board). It can be from 4 s before to 3 s after the start of the wave.
2. **Still hands before:** right before the push (at most 0.2 s of rise), the wrist below 60 °/s (for about 0.28 s or more). The pop-up starts where that stillness starts, that is, where the last stroke ended.
3. **A stroke in the preceding second** (over 120 °/s): if you weren't paddling it isn't a take-off.
4. **Standing:** after the push peak, the first moment the wrist stays under 100 °/s for 0.32 s. That is where the pop-up ends.
5. Between 0.32 s and 3 s; if more than one push qualifies, the strongest.

Without a gyroscope, or if the pause of the hands doesn't show up, there is no pop-up ("—" on the watch, 0 in Garmin Connect). The thresholds come from the expected movement, not from wrist measurements. To tune them with your sessions in Calibrate:

1. `tools/rawlog.py takeoffs <activity>.fit <folder>` builds one chart per take-off with the rule marked, and a `takeoffs.csv`.
2. In `takeoffs.csv` you note the actual pop-up of each wave in seconds (from the video, or by looking at the chart), or "-" if there was none.
3. `tools/popup.py tune <folder>/takeoffs.csv ...` finds the thresholds with the least error, and with `--apply` writes them into both twins and rebuilds the test vectors, the manual's figure and the list above (generated from the thresholds: `tools/popup.py docs`).

On synthetic take-offs of known duration (`tools/popup.py tune --synthetic`), the current rule is off by 0.03 s on average and tuning brings it down to 0.01 s. On synthetic take-offs built from the phases below, the rule gives about 0.08 s less than what was built (the smoothing eats away the edges).

What the studies say (all on land, with a force platform; there are no confirmed published numbers for in the water, nor for beginners):

Who	From → to	Time	Source
Competitive surfers, men (9)	chest lifts off → front foot lands	0.56 ± 0.06 s	Parsonage et al. 2018
Competitive surfers, women (8)	Same	0.68 ± 0.07 s	Same
Experienced recreational surfers (23 men)	hands touch the board → weight stable	1.20 ± 0.19 s	Borgonovo-Santos et al. 2021
Intermediates, ~25 years old	Same	1.12 ± 0.16 s	Enticott 2024 (thesis)
Intermediates, ~52 years old	Same	2.03 ± 0.52 s	Same

The watch measures the second kind (from hands on the board to being stable), which includes the landing. Borgonovo-Santos splits it into pushing with the hands on the board 0.71 ± 0.08 s, transition ~0 s and from landing to being stable 0.48 ± 0.22 s; the test take-offs are built with those phases (`tools/popup.py`). No study publishes what the pop-up looks like at the wrist (g or °/s): those that use body-worn sensors put them on the back, the sacrum or the legs.

Known issues

Symptom	Fix
`Unable to locate a Java Runtime`	`brew install openjdk@17` (the scripts already add it to the PATH)
`no devices found` when uploading	Watch in USB **MTP** mode, data cable, `killall ptpcamerad`
PTP error 2002	Use `scripts/deploy.sh` (ciqsend), not `mtp-sendfile`
Grey values, "N H AGO"	There has been no data for over 3 h: the phone has no connection or Garmin Connect is closed. If there is also a crossed-out phone, the watch isn't connected to the phone
Some text doesn't show up	The character is missing from `SPECS` in `make_fonts.py`
Changes don't show on the watch	You have to unplug the watch so it installs
I changed spot and it stays on "—" / LOADING	Spots within 25 km come preloaded; a farther one waits for the next background request (Garmin allows one every 5 min; the face retries every 6). If they don't arrive, check that there isn't a crossed-out phone

`make_dial.py` uses the macOS system font (`SFNS.ttf`) for N/E/S/W. On another system, change the path to another font, for example `tools/fonts/Barlow-SemiBold.ttf`.